

Jake Doherty

Austin, TX

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Education

University of Texas at Austin

2022–Present

PhD Candidate, Physics

University of South Florida

May 2022

B.S. Physics, B.A. Mathematics

Technical Skills

Programming: Python, C++, C, Fortran, Bash

Scientific Computing: NumPy, SciPy, Matplotlib, Linux/Unix, Slurm, Git

Methods: Numerical integrations of differential equations, Monte Carlo methods, Statistical modeling, Machine learning algorithms

High Performance Computing: Parallel and cluster computing, Large-scale numerical simulations

Research Experience

Graduate Researcher, University of Texas at Austin

Aug 2022–Present

Advisor: Pablo Laguna

- Design and implement high-order finite-difference algorithms for solving coupled nonlinear partial differential equations using C++ and Fortran.
- Run large-scale simulations on HPC clusters using distributed computing resources.
- Analyze very large simulation datasets and build automated post-processing pipelines using Bash scripts and Python.

Research Assistant, University of South Florida

May 2020–Aug 2022

Advisor: Inna Ponomareva

- Modeled ferroelectric and antiferroelectric materials using computational physics methods.
- Performed numerical analysis of polarization, phase transitions, and electronic structure.
- Developed Python workflows for analysis and visualization of large datasets.

Selected Projects

Black Hole Growth in Scalar Field Cosmology

- Performed large scale numerical integration of PDEs on HPC clusters to study black hole spacetimes.
- Implemented numerical algorithms for manipulation of output data, including interpolation and plotting.

Numerical Relativity Infrastructure

- Extended Einstein Toolkit infrastructure with custom numerical solvers and gauge conditions.
- Implemented high-order interpolation, adaptive mesh refinement workflows, and parallel analysis tools.

Publications

Physical Review D 112, 044029 (2025)

Growth of a Black Hole in a Scalar Field Cosmology

Journal of Physical Chemistry C 125(29), 16296–16303 (2021)

Tunability of Structure, Polarization, and Band Gap of High- T_c Organic-Inorganic Ferroelectrics by Hydrostatic Pressure

Journal of Applied Physics 132, 034101 (2022)

Negative Capacitance Regime in Antiferroelectric PbZrO_3

Leadership & Teaching

Teaching Assistant, University of Texas at Austin	2022–Present
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Directed Reading Program Mentor, University of Texas at Austin	2022–Present
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Teaching Assistant, University of South Florida	2021–2022
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Awards

USF Dean’s List of Scholars (2021)

USF Director’s Award (2018)

Florida Academic Scholars Award (2018)